

Low-Level System Design

Version: 1.0

Document Control

Version	Date	Author	Description
1.0	2026-05-12	Hardik Patel	Initial Version

Reviewers

Name	Role	Approval Date
Hardik Patel	CTO	2026-05-13

Approvers

Name	Role	Approval Date
Hardik Patel	CTO	2026-05-14

Module Overview

Provide a summary of the module or service.

Include:

- Module purpose
- Business capability
- Key responsibilities

Example:

The Order Service manages order creation, order status updates, payment validation, and order lifecycle tracking.

Responsibilities

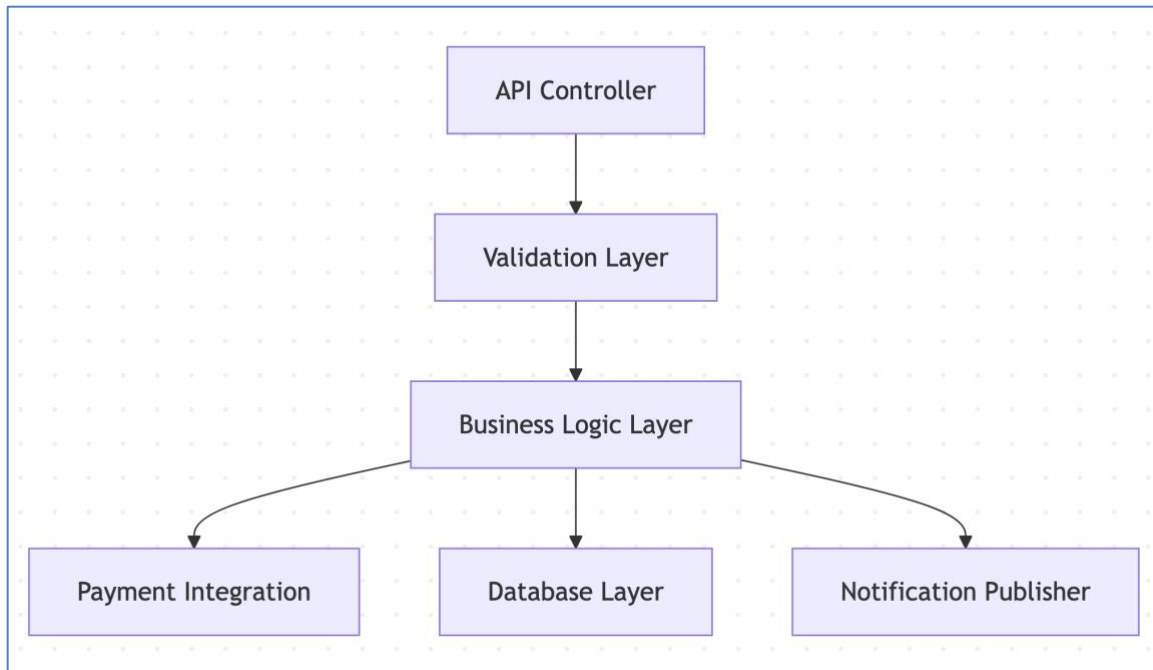
Define major responsibilities of the module.

Responsibility ID	Description
RESP-001	Validate order requests
RESP-002	Create customer orders
RESP-003	Coordinate payment processing
RESP-004	Update order status
RESP-005	Publish order notifications

Component Architecture

Describe internal components/modules.

Example Diagram:



API Endpoints

List APIs exposed by this module.

API	Method	Description
/orders	POST	Create order
/orders/{id}	GET	Retrieve order
/orders/{id}/cancel	POST	Cancel order

Request Validation Rules

Define request validation logic.

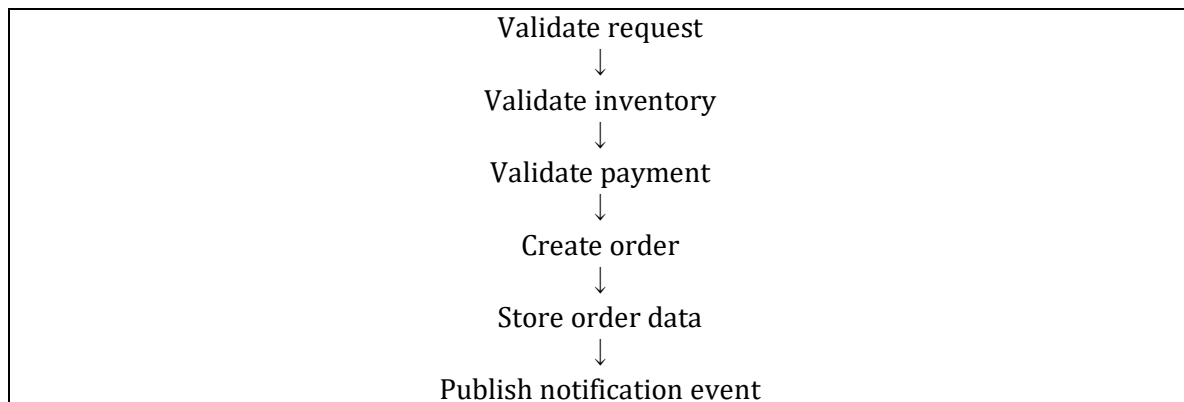
Field	Validation Rule
userId	Required
productId	Must exist in product catalog
quantity	Must be greater than zero

paymentMethod	Supported payment method required
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Business Logic Flow

Describe major business rules.

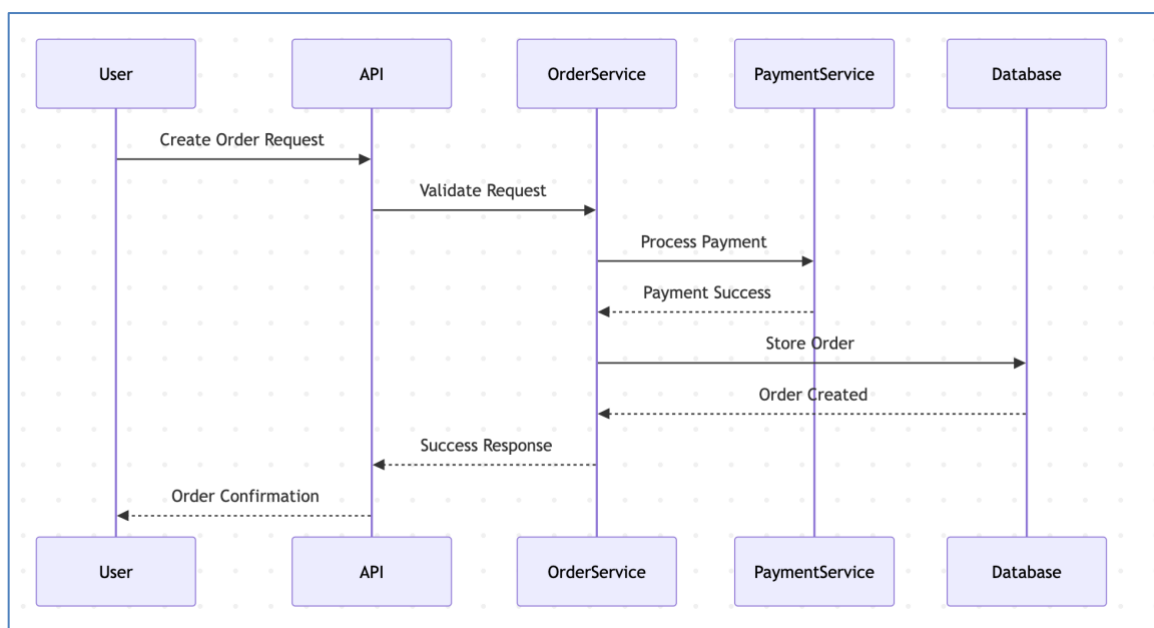
Example:



Sequence Diagram

Describe request lifecycle.

Example Sequence:



Database Interactions

Describe database operations.

Operation	Entity	Purpose
INSERT	Orders	Create new order
UPDATE	Orders	Update order status
INSERT	Payments	Store payment details
SELECT	Inventory	Validate stock availability

Caching Strategy

Document caching behavior if applicable.

Area	Strategy
Product Catalog	Redis cache
User Sessions	In-memory/session cache
Frequently Accessed Orders	Short-term cache

Error Handling

Define expected failure scenarios.

Scenario	Handling
Payment Failure	Return payment retry message
Product Out of Stock	Prevent order creation
Database Timeout	Retry operation
Invalid Authentication	Return 401 Unauthorized

Retry & Resilience Strategy

Define retry logic and resilience handling.

Area	Strategy
Payment APIs	Retry with exponential backoff
Notifications	Queue-based retries
External APIs	Circuit breaker pattern

Logging & Monitoring

Define observability expectations.

Area	Requirement
Request Logs	Log all incoming requests
Error Logs	Log failures with correlation IDs
Metrics	Capture API latency and error rates
Monitoring	Alerts for service degradation

Security Controls

Document security expectations.

Area	Requirement
Authentication	JWT validation
Authorization	RBAC checks
Encryption	Sensitive data encrypted
Input Validation	Prevent malicious requests

Rate Limiting & Abuse Prevention

Document protection mechanisms.

Area	Strategy
API Requests	Rate limiting
Login Attempts	Brute-force prevention
Public APIs	Request throttling

Performance Considerations

Describe optimization strategies.

Area	Optimization
Database Queries	Indexed queries
API Responses	Pagination
External Calls	Async processing
Heavy Operations	Background jobs

Dependencies

Document module dependencies.

Dependency	Purpose
Payment Service	Payment processing
Inventory Service	Product availability
Notification Service	Customer notifications

Assumptions

Document implementation assumptions.

Example:

- Inventory service provides near real-time stock updates
- Payment gateway APIs are highly available
- Notification service supports retry handling

Constraints

Document known technical limitations.

Constraint Type	Description
Legacy Systems	Existing payment provider required
Compliance	PCI compliance required
Performance	API latency under 500ms

Risks

Identify implementation risks.

Risk ID	Description	Impact
RISK-001	Inventory synchronization delays	HIGH
RISK-002	Payment provider instability	HIGH
RISK-003	High API traffic spikes	MEDIUM

Test Considerations

Define testing expectations.

Test Type	Scope
Unit Testing	Business logic validation
Integration Testing	External service integration

API Testing	Endpoint validation
Performance Testing	Load handling verification

Open Questions

Track unresolved engineering questions.

Question ID	Question	Owner
Q-001	Should order creation be synchronous or async?	Engineering Team
Q-002	What retry limit should payment APIs use?	Architecture Team

Related Documents

Document	Purpose
BRD	Defines business requirements
PRD	Defines product requirements
Feature Specification	Feature Behavior
API Specification	API contract definitions
DB Schema	Data model definitions
High-Level Design (HLD)	System-wide architecture